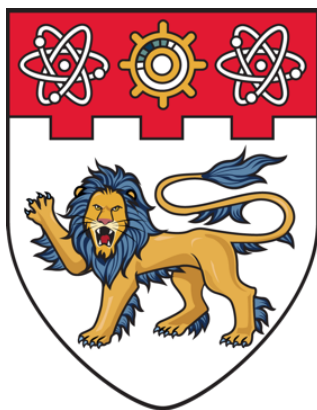




**NANYANG
TECHNOLOGICAL
UNIVERSITY**

SINGAPORE

**AI6127 DEEP NEURAL NETWORKS FOR NATURAL
LANGUAGE PROCESSING**

**FINAL PROJECT REPORT ON:
EXPLORING THE MULTILINGUAL DIALOGUE GENERATION TASK**

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Abstract

Multilingual dialogue generation is a crucial sequence-to-sequence task for developing cross-lingual conversational AI, enabling global communication and supporting low-resource languages. Before the rise of Large Language Models, existing multilingual models primarily focused on machine translation, lacking dialogue-specific reasoning and struggling with context understanding and data imbalance. To explore this task, we fine-tune and evaluate five pre-LLM multilingual models, including mBART, mT5, and M2M100, using the English-Chinese bilingual subset of the XDailyDialog corpus. We employ a multi-dimensional assessment framework, including accuracy metrics (BLEU, ROUGE-L) and diversity metrics (Distinct-N). Our key finding is that the M2M100-1.2B model achieves the highest scores across all similarity metrics (BLEU: 5.089, ROUGE-L: 10.321), demonstrating the best contextual understanding. Through the analysis of the results, we found that the Cross-lingual Interference phenomenon, the scale of model parameters, and the high flexibility of the conversational dataset limit the effectiveness of some models. Furthermore, we confirm that larger models generally yield better results, and demonstrate that hyperparameter tuning, such as adding generation length and n-gram repetition penalties, is crucial for mitigating overly safe or repetitive outputs.

1 Introduction

The Multilingual Dialogue Generation (MDG) task is essentially a sequence-to-sequence (seq2seq) generation task. It requires a model to take a dialogue context as input and generate the next utterance as output. The significance of MDG is profound, as it is crucial for building cross-lingual conversational AI, including multilingual chatbots and cross-lingual assistants. This capability is vital for enabling global communication and providing necessary support for low-resource languages. Unlike mere translation, MDG focuses on producing contextual, human-like responses across multiple languages.

However, the task presents several inherent difficulties. Before the rise of Large Language Models (LLMs), earlier multilingual models such as mBART [1], mT5 [2], and M2M100 [3] primarily focused on general text generation or machine translation. While effective in language transfer, these models lacked the dialogue-specific reasoning and context tracking necessary for conversational systems. Furthermore, dialogue-specific multilingual datasets were often limited or unbalanced. Key challenges persist, including difficulties in context understanding across multiple languages, data imbalance among them, and the complexity of handling code-switching and emotionally coherent responses. Most of these models were not originally fine-tuned for dialogue but were only adapted from general-purpose language models.

To address these challenges and explore the performance of pre-LLM multilingual models on the MDG task, our approach involves fine-tuning the mBART (mBART25, mBART50), mT5-base, and M2M100 (M2M100-418M, M2M100-1.2B) architectures. We utilized the XDailyDialog dataset [4], which is an extended multilingual parallel version of the popular Daily Dialog corpus. Specifically, we selected the English-Chinese bilingual subset, which contains conversational data aligned at the utterance level.

To thoroughly evaluate the models, we employed a multi-dimensional assessment framework. This framework combined accuracy metrics, specifically BLEU and ROUGE, to assess content quality and similarity to human references, with diversity metrics, Distinct-N (distinct 1 and distinct 2), to measure conversational richness and prevent generic outputs. Finally, human evaluation was conducted to assess actual quality, fluency, relevance, and coherence beyond surface-level text overlap.

2 Related Works

Prior to the dominance of large language models (LLMs), multilingual NLP was primarily developed through two main architectural families, focused on either deep representation learning or sequence-to-sequence generation.

Early attempts at cross-lingual sequence-to-sequence tasks relied on machine translation (MT)[5] [6], either by translating the test set into the source language or by translating the training data into the target language. This MT-based approach has the notable advantage of being less susceptible to the "curse of multilinguality"[7] often encountered in large-scale multilingual encoders. However, its effectiveness is severely constrained in task coverage because the development and quality of MT models are heavily dependent on the availability of suitable corpora.

The advent of the encoder-decoder architecture, however, transformed cross-lingual tasks into a general, flexible, and high-performing paradigm capable of learning shared cross-lingual representations within a unified framework, which can be applied to a wider range of Seq2Seq tasks such as summarization, dialogue systems, and semantic parsing.

Encoder-Only Models focused on creating robust, shared contextual representations across diverse languages. Models such as mBERT [8] and its successor, XLM-R [9], achieved strong cross-lingual transfer capabilities by being pre-trained on massive, shared multilingual corpora. While foundational for language understanding, these models face unique challenges in generation tasks, particularly in dialogue: they struggle with contextual coherence and accurately handling code-switching, a key issue in real-world multilingual communication. To mitigate these issues, prior research explored techniques such as *Simulating Code-Switching* via data augmentation [10] and various methods for *Tuning the Multilingual Space*, including decoupling language and task knowledge through knowledge distillation [11].

Encoder-Decoder Multilingual Architectures directly targeted the sequence-to-sequence generation problem, forming the immediate benchmark for multilingual dialogue generation:

- **mBART**: This was an early, prominent seq2seq Transformer that pioneered multilingual denoising pre-training [1].
- **mT5**: A seq2seq model that scaled the T5 framework to multilingual data using the massive Multilingual C4 dataset [2].
- **M2M100**: Designed specifically for Multilingual Machine Translation, this model introduced techniques for non-English centric translation paths across 100 languages [3].

These encoder-decoder models established the performance frontier for general multilingual generation, including translation and summarization, serving as the essential background for our evaluation of multilingual dialogue systems.

3 Method

3.1 mBART

The mBART (Multilingual BART) architecture is a sequence-to-sequence (seq2seq) Transformer model. It follows a standard Encoder-Decoder design, similar to the original BART and T5 models. mBART was pre-trained on large-scale multilingual corpora, utilizing a Denoising Autoencoding objective [1]. This objective requires the model to reconstruct the original text from corrupted multilingual input during pre-training.

In our study, we evaluated two variants of the architecture: mBART-25, which was trained on 25 languages, and mBART-50, trained on 50 languages. The mBART framework supports multilingual learning through the use of language tags (e.g., <En>, <Ja>), which explicitly specify the source and target languages. Importantly, the same architecture is versatile enough to support both sentence-level (Sent MT) and document-level (Doc MT) generation tasks.

3.2 M2M-100

M2M-100 is a multilingual translation model launched by Facebook AI in 2020. Its full name is "Many-to-Many 100". The name encapsulates its core capability: it can perform direct "many-to-many" translation

between 100 different languages. Unlike many previous models, it does not rely on English as an intermediary language. It can translate directly between the two languages.

The M2M-100 skeleton is a standard Transformer model, but a model itself does not know the specific language of a sentence. To address this issue, researchers assigned a Language ID token to each language, for example, $[zh]$ for Chinese and $[fr]$ for French. When you need to translate a sentence, you must add the source language’s Language ID token at the beginning of the sentence. This ID token tells the encoder which language it is reading next, and tells the decoder which language it expects to generate. In this way, M2M-100 can achieve 100×99 translation directions using only one model.

The M2M-100 adopts a parameter sharing strategy. 100 languages share the same set of encoder and decoder parameters. This means that the knowledge that the model acquires when learning language 1 may also be used to help understand language 2, which has a similar structure. This is crucial for improving the translation quality of languages with scarce data.

M2M-100 uses the SentencePiece word segmentation algorithm to create a unified, shared vocabulary for all 100 languages. This strengthens the connections between languages.

These designs transform a single Transformer model into a powerful multilingual translation system.

3.3 mT5

mT5 is a multilingual extension of the T5 model, retaining its central design philosophy: the unified “text-to-text” format [12]. This structure merges all Natural Language Processing tasks, such as classification, question answering, and generation, into a format where the model receives text as input and produces text as output. Architecturally, mT5 utilizes a standard Encoder-Decoder Transformer setup, which is particularly robust for handling complex sequence transformation tasks, including machine translation and summarization [2].

The core strength of mT5 lies in its extensive multilingual pre-training. The model was trained on a novel dataset called mC4 (multilingual C4) [2], which is an expanded version of the original English C4 (“Colossal Clean Crawled Corpus”) dataset [12]. The mC4 dataset includes textual data from 101 distinct languages, allowing the model to learn powerful cross-lingual representations. The pre-training objective follows the original T5 design, utilizing a span-masking denoising task where the model is trained to reconstruct missing input.

The advantage of the mT5 model lies in its zero-shot and few-shot learning capabilities, making it particularly suitable for multilingual tasks that lack large amounts of labeled data[2].

4 Experiments

4.1 Data

Our project utilizes the XDailyDialog dataset [4], which is an extended multilingual parallel version of the popular Daily Dialog corpus. This resource provides dialogue pairs translated into multiple languages, making it suitable for both multilingual and cross-lingual generation tasks. We specifically selected the English-Chinese bilingual subset, which features human-translated conversational data aligned at the utterance level.

The associated task is Multilingual Dialogue Generation, which is modeled as a sequence-to-sequence task where the model takes a dialogue context as input and generates the next utterance as output.

Input and Output Format Each training sample is structured as a source utterance (context) and a target utterance (response). These can optionally be across different languages. To guide the generation process, several control tokens are prepended to the input sequence:

- **Language Tags:** Tokens like <en> and <zh> are added to indicate the required source and target languages for the generation.
- **Dialogue Metadata:** Information concerning the dialogue’s topic, action, and emotion is included. This metadata helps the model identify the overall dialogue structure.

For example, in an English to English task, the input format includes metadata, the context, and the target language tag (<En>). Similarly, for a Chinese to Chinese task, the input utilizes the corresponding Chinese tag (<Zh>) to specify the generation language.

4.2 Evaluation method

4.2.1 BLEU score

BLEU (Bilingual Evaluation Understudy)[13] is an automated metric widely used to evaluate the quality of machine translation. BLEU is not just a single fraction. It also relies on the precision calculation of smaller units, namely "n-grams". We primarily used two metrics: BLEU-1 and BLEU-2.

BLEU-1 primarily measures the matching accuracy of individual words. This primarily reflects the translation’s performance in terms of lexical accuracy, i.e., whether the correct words were used.

BLEU-2 measures the matching accuracy of phrases consisting of two words. It reflects, to some extent, the fluency of a phrase, that is, whether the collocation between words is reasonable.

Applications in dialogue systems. Although BLEU originated from machine translation, it is also widely used to evaluate the response quality of generative dialogue systems. [14] For example, it can compare generated responses with human references and evaluate translation quality in multilingual dialogues.

Limitation. BLEU only cares about the overlap of words and phrases, not semantic equivalence or contextual coherence. Therefore, manual evaluation is essential.

4.2.2 ROUGE score

ROUGE (Recall-Oriented Understudy for Gisting Evaluation)[15] was originally designed as a metric to evaluate the quality of automatic text summarization. Its core idea is to calculate the degree of overlap between machine-generated text and human-written reference text.

We used three metrics: ROUGE-1, ROUGE-2, and ROUGE-L. ROUGE-1 measures the overlap of individual words, primarily reflecting the degree of information coverage. ROUGE-2 measures the overlap between two consecutive words, better reflecting the fluency of the text and the accuracy of phrase collocation. ROUGE-L measures the longest common word sequence between two text sequences, capturing sentence-level structural similarity.

Application in Dialogue Systems. Although ROUGE originated from the summarizing task, in dialogue systems, it can be used to evaluate the completeness of the response. It can assess whether the system-generated responses cover key information. It measures how much relevant and necessary content is included. If a model’s response is fluent but misses the key information the user asked, its ROUGE score may be low.

4.2.3 Distinct-N

Distinct-N [16] is an automated metric used to evaluate the diversity of text generation. It is widely employed in dialogue generation tasks to quantify the richness and repetitiveness of model responses. The core mechanism of this metric is to calculate the proportion of unique n-grams (sequences of N consecutive words) in all the texts generated by the model to the total number of n-grams.

In our experiment, we used the Distinct-1 and Distinct-2 indicators for evaluation. Distinct-1 measures lexical diversity (i.e., the number of unique words used), while Distinct-2 measures the diversity of longer phrases. A higher Distinct-N score (the closer to 1.0, the better) indicates that the model-generated responses

are rich, with varied word usage and sentence structure combinations, and rarely self-replicate; conversely, a lower score implies that the model’s output is very monotonous and repetitive, or tends to frequently use common ”safe responses” such as ”I don’t know” and ”OK”.

4.2.4 Manual Inspection

In the dialogue generation task, Manual Inspection serves as the gold standard for evaluating the performance of the model. This method relies entirely on trained human evaluators to thoroughly read, analyze, and judge the authenticity of the responses generated by the model. It overcomes the limitations of automatic metrics such as BLEU or ROUGE because dialogues are inherently open-ended, and a ”good” response has multiple possibilities, while automatic metrics often fail to accurately capture the semantic and pragmatic quality at these levels [17]. During the manual evaluation process, the evaluators use a detailed set of evaluation criteria to comprehensively assess the responses from multiple key dimensions. In our experiments, we simply adopted a sampling comparison approach to determine the superiority or inferiority of different models’ responses to the same set of dialogues.

4.3 Model settings

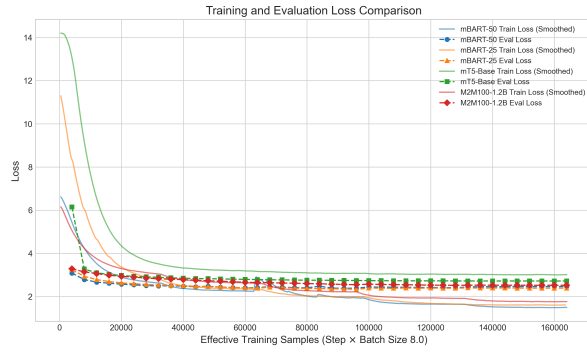
In order to better compare the individual performance of the models, after a series of tests, we adopted the same set of hyperparameters for training. Some important hyperparameters are shown in Table 1. All training was conducted on RTX 5090, T4 or A100 GPUs.

Table 1: Uniform hyperparameter setting

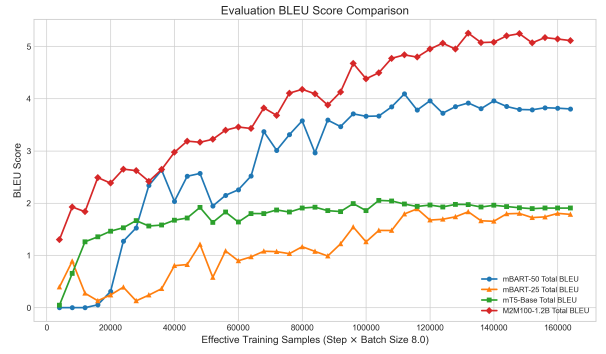
Hyperparameter	Value
num_beams	3
epochs	5
batch size	8 * 2 (Total 16)
learning_rate	2e-5
length_penalty	1.5
no-repeat_ngram_size	3

4.4 Results

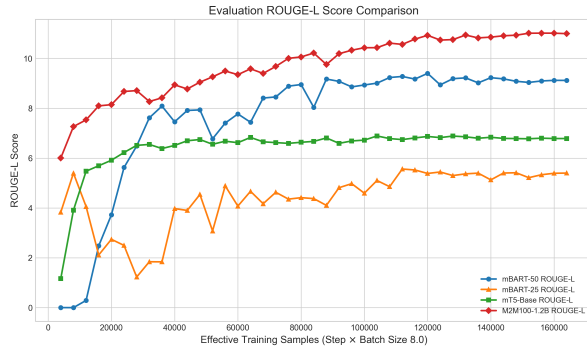
All models were trained for 5 epochs, the statistical metric curves during the training process are shown in Figure 1 and the final results of the model on the test set are shown in Table 2.



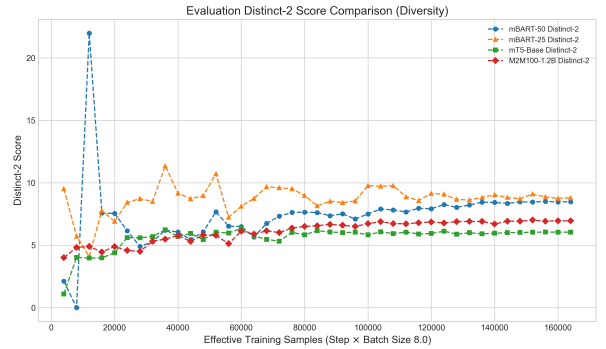
a Comparison of loss curves during training



b Comparison of model performance (BLEU) during training



c Comparison of model performance (ROUGE-L) during training



d Comparison of model performance (Distinct-2) during training

Figure 1: Comparison of model performance metrics during training

In terms of training stability, all models showed a similar trend in loss, though the mT5-base was characterized by an extremely high initial loss, followed by an eventual stagnation in performance.

Across similarity-based metrics, the M2M100-1.2B model achieved the highest scores for BLEU and ROUGE-L, confirming its ability to generate responses that are similar to human-written responses. Furthermore, compared to M2M models of the same category, we can find that models with a larger number of parameters generally yielded better performance. In addition, a manual review of the generated output corroborated this, showing that the M2M100 model demonstrated the best performance in terms of contextual understanding and grammatical logic.

While the M2M100 models excelled in BLEU and ROUGE scores, they exhibited the lowest scores in the diversity metrics Distinct-1 and Distinct-2, suggesting a tendency toward more generic output. In contrast, the mBART family demonstrated higher diversity scores, with mBART-25 achieving the highest Distinct-2 score of 9.075. However, mBART-25 also yielded the worst generation quality metrics (BLEU: 1.772, ROUGE-L: 5.494). The mBART-50 model, though, ranking second in overall generative quality, represented a balanced option.

Table 2: Model Performance Metrics on Test Set

Model	Loss	BLEU	BLEU-1	BLEU-2	ROUGE-L	Distinct-1	Distinct-2
m-BART25	2.398	1.772	27.375	4.655	5.494	1.434	9.075
m-BART50	2.455	3.748	28.652	6.381	8.822	1.199	8.883
M2M100-1.2B	2.542	5.089	29.354	7.372	10.321	0.963	7.173
M2M100-418M	2.822	3.494	28.307	5.907	9.176	0.963	6.405
MT5-base	2.730	2.173	23.864	4.455	6.999	1.016	5.985

After human evaluation of the generated texts, we concluded that the M2M-100 model’s results were superior to mBART, which were in turn superior to mT5-base. Furthermore, within the same class of models,

those with larger parameter counts yielded better generation performance.

*** Sample (Original Index: 1427) ***

Source :

5 1 0 [SEP] Brian , do you know how to speak English ?

Target :

Yes .

[MT5BASE Prediction]: No , I don't know how to speak English

[M2M1001_2B Prediction]: Not very well . I'm not very good at speaking English .

[M2M100_418M Prediction]: Yes , I know how to speak English

[MBART50 Prediction]: Yes , I know how to speak English . I have been studying English for three years .

[MBART25 Prediction]: , I'm not very good at it.

a An example of short English dialogue

*** Sample (Original Index: 2014) ***

Source :

5 1 0 [SEP] hi , Mary ! What do you like to do in your spare time ?

[SEP] well , I spend a lot of time watching movies .

[SEP] what a confidence ! I always watch a lot of movies , too .

[SEP] oh really , Frank ? What kind of movies do you like ?

[SEP] actually , I watch whichever movie there is , be it a comedy , a sci-fi or a suspense

movie . How about you ?

[SEP] art films are my favorite , but thrillers are cool , too .

[SEP] really impressive . These two genres are totally different .

[SEP] sure . I enjoy both of them very much .

[SEP] how often do you go to the cinema ?

[SEP] once in a while , I suppose , I usually rent movies at Movie Salon .

[SEP] movie salon ? where is that ?

Target :

it's a movie rental store in my neighborhood . I've got a membership there .

[MT5BASE Prediction]: actually , there is a movie salon , but I don't like it .

[M2M1001_2B Prediction]: it's just around the corner . There's one on the left .

[M2M100_418M Prediction]: it's on the first floor of the cinema .

[MBART50 Prediction]: I'm at the cinema on the corner of 19th street . it's right behind

my house .

[MBART25 Prediction]: 哦 , it's on the first floor , on the third floor.

c An example of long English dialogue

*** Sample (Original Index: 3288) ***

Source :

1 1 0 [SEP] 我要了。多少钱？ <Zh>

Target :

60美元。

[MT5BASE Prediction]: 你需要多少钱?

[M2M1001_2B Prediction]: 请稍等,先生。我们的价格是12美元。

[M2M100_418M Prediction]: 嗯,这是您的收据。

[MBART50 Prediction]: 每箱120元。

[MBART25 Prediction]: 每打二十美元。

b An example of short Chinese dialogue

*** Sample 16 (Original Index: 4474) ***

Source :

5 1 0 [SEP] 莎拉 , 你今天做了什么?

[SEP] 我去购物了。

[SEP] 你买了什么吗?

[SEP] 是的 , 我买了一些东西。

[SEP] 买了什么?

[SEP] 这件外套。你喜欢吗?

[SEP] 我很喜欢 , 非常漂亮。你在哪里买的? <Zh>

Target :

在第五大街的购物中心。

[MT5BASE Prediction]: 我买的。

[M2M1001_2B Prediction]: 我在百货公司买的。

[M2M100_418M Prediction]: 我去购物中心买的。

[MBART50 Prediction]: 我在宜家买的。

[MBART25 Prediction]: 我去宜家买的。

d An example of long Chinese dialogue

Figure 2: Examples of Model Generation Output

5 Analysis

Further analysis through human evaluation revealed the reason why the mBART-25 exhibiting high diversity but low quality—the generated result occasionally exhibited language confusion. In some specific topics, the input language was English, yet the model would suddenly generate Hindi characters. This phenomenon is a classic example of Cross-lingual Interference [18], inherent to multilingual pre-trained Seq2Seq models. The phenomenon we observed in the experiments may indicate that high-resource languages like Hindi might over-influence the attention mechanisms, causing it to incorrectly activate Hindi-related generation paths instead of producing normal English replies.

The mT5-base model struggled differently, it tends to generate outputs that are overly structured and syntactically rigid, and sometimes nonsensical responses. Its tendency to output safe sentences with low lexical diversity points to the performance bottleneck observed in the training process. Given its relatively smaller parameter count compared to its larger siblings or concurrent models, the mT5-base model lacks the capacity required to store and retrieve extensive common sense and general world knowledge.

Furthermore, the complexity of the XDailyDialog dataset—a multi-turn, open-domain corpus was identified as a major challenge for the pre-LLM models examined in this experiment. The difficulty stems not only from its requirement for conversational coherence over multiple turns, but also from the broad and unrestricted nature of topics. In particular, in certain dialogues, the corpus even exhibited highly nuanced and

context-dependent sarcastic responses (Figure 3), which is a sophisticated communicative act that proved exceptionally difficult for the models to learn and replicate accurately.

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*** Sample (Original Index: 520) ***
Source :
1 4 4 [SEP] I am so busy today .
[SEP] Can I help you ?
[SEP] I ' m not sure . Perhaps you could do the shopping .
[SEP] I ' ll be glad to help .
[SEP] Or maybe make the beds .
Target :
All you have to do is ask for help .
-----
[MT5BASE Prediction]: Yes , I ' m going to make the beds .
[M2M1001_2B Prediction]: That would be very kind of you .
[M2M100_418M Prediction]: That ' s great . I ' ll be glad to help .
[MBART50 Prediction]: I ll be glad to help you .
[MBART25 Prediction]: thanks.

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Figure 3: An example of sarcastic response

Despite these challenges, the experiments yielded valuable insights regarding hyperparameter tuning. First, the introduction of a generation length penalty proved necessary to counter the model’s tendency to produce overly safe and concise outputs, encouraging longer, more informative responses. Second, applying an N-gram repetition penalty was crucial for controlling the word and phrase redundancy that occurs during Beam Search. These two measures were found to significantly improve the fluency and readability of the final dialogue generated by models.

6 Conclusion

Through our previous experiments, we confirmed that pre-LLM models such as M2M-100 possess good conversational generation capabilities in bilingual dialogue tasks. However, it is undeniable that the task of dialogue generation, which is diverse and involves highly flexible language use, remains extremely difficult for model training and result evaluation, especially for models that were not specifically designed for this task. These findings collectively highlight that while these Seq2Seq models are powerful for translation, they still have fundamental shortcomings in contextual understanding and knowledge application when handling complex, multi-turn dialogue dynamics.

Naturally, the current experiment has notable limitations. For instance, the relatively small scale and limited corpus of the XDailyDialog dataset itself may have constrained the model’s ability to learn robust knowledge for the dialogue task. Moreover, the current task only incorporated Chinese and English as training inputs, neglecting to evaluate the model’s efficacy in broader multilingual contexts. Subsequent research could potentially be enhanced by conducting experiments with datasets that feature a richer diversity of languages and a larger corpus.

7 Acknowledgments

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